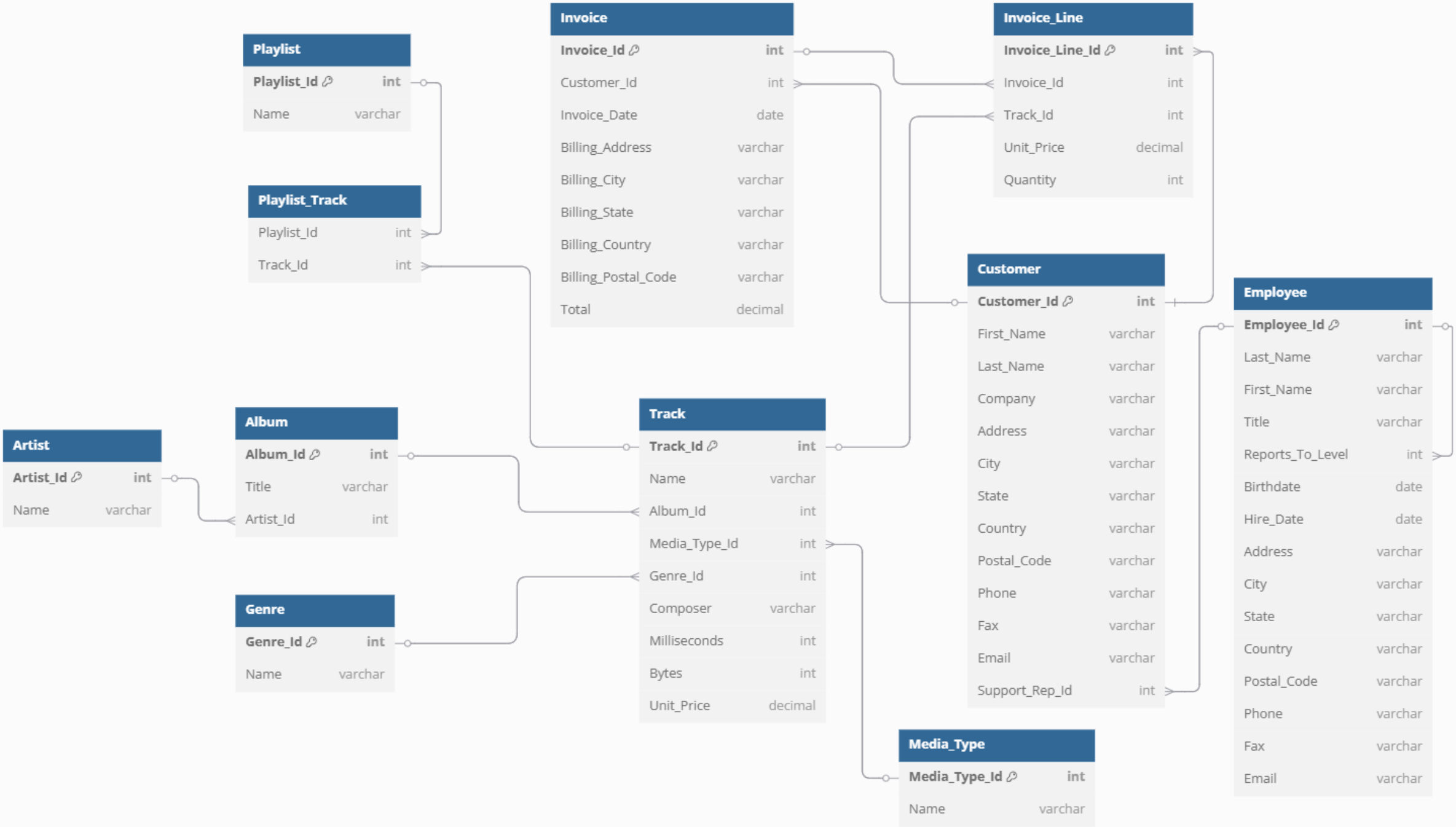
MUSIC STORE

SQL Database


Playlist
Playlist_Id int
Name varchar
Playlist_Track
Playlist_Id int
Track_Id int
Invoice
Invoice_Id int
Customer_Id int
Invoice_Date date
Billing_Address varchar
Billing_City varchar
Billing_State varchar
Billing_Country varchar
Billing_Postal_Code varchar
Total decimal
Invoice_Line
Invoice_Line_Id int
Invoice_Id int
Track_Id int
Unit_Price decimal
Quantity int
Customer
Customer_Id int
First_Name varchar
Last_Name varchar
Company varchar
Address varchar
City varchar
State varchar
Country varchar
Postal_Code varchar
Phone varchar
Fax varchar
Email varchar
Support_Rep_Id int
Employee
Employee_Id int
Last_Name varchar
First_Name varchar
Title varchar
Reports_To_Level int
Birthdate date
Hire_Date date
Address varchar
City varchar
State varchar
Country varchar
Postal_Code varchar
Phone varchar
Fax varchar
Email varchar
Artist
Artist_Id int
Name varchar
Album
Album_Id int
Title varchar
Artist_Id int
Track
Track_Id int
Name varchar
Album_Id int
Media_Type_Id int
Genre_Id int
Composer varchar
Milliseconds int
Bytes int
Unit_Price decimal
Genre
Genre_Id int
Name varchar
Media_Type
Media_Type_Id int
Name varchar

1. WHO IS THE SENIOR MOST EMPLOYEE BASED ON JOB TITLE?

```
SELECT
    *
FROM
    employee
ORDER BY LEvel DESC
LIMIT 1;
```

employee_id	last_name	first_name	title	reports_to	level	birthdate	hire_date
1	Adams	Andrew	General Manager	9	L6	18-02-1962	14-08-2016

2. WHICH COUNTRIES HAVE THE MOST INVOICES?

```sql
SELECT
    billing_country, COUNT(*) AS Total_invoices
FROM
    invoice
GROUP BY billing_country
ORDER BY TOtal_invoices DESC;
```

billing_country	Total_invoices
USA	131
Canada	76
Brazil	61
France	50
Germany	41
Czech Republic	30

3. WHAT ARE TOP 3 VALUES OF TOTAL INVOICE?

```sql
SELECT
    Total
FROM
    invoice
ORDER BY Total DESC
LIMIT 3
```

Total
23.759999999999998
19.8
19.8

4. WHICH CITY HAS THE BEST CUSTOMERS? WE WOULD LIKE TO THROW A PROMOTIONAL MUSIC FESTIVAL IN THE CITY WE MADE THE MOST MONEY. WRITE A QUERY THAT RETURNS ONE CITY THAT HAS THE HIGHEST SUM OF INVOICE TOTALS. RETURN BOTH THE CITY NAME & SUM OF ALL INVOICE TOTALS

```sql
SELECT
    billing_city, SUM(total) AS TOtal_sum
FROM
    invoice
GROUP BY billing_city
ORDER BY TOtal_sum DESC
LIMIT 1
```

billing_city	Total_sum
Prague	273.24000000000007

5. WHO IS THE BEST CUSTOMER? THE CUSTOMER WHO HAS SPENT THE MOST MONEY WILL BE DECLARED THE BEST CUSTOMER. WRITE A QUERY THAT RETURNS THE PERSON WHO HAS SPENT THE MOST MONEY

```sql
SELECT
    customer.first_name,
    customer.last_name,
    customer.customer_id,
    SUM(Total) AS Total_sum
FROM
    Customer
        JOIN
    invoice ON customer.customer_id = invoice.invoice_id
GROUP BY customer.customer_id , customer.first_name , customer.last_name
ORDER BY Total_sum DESC
LIMIT 1
```

first_name	last_name	customer_id	Total_sum
Martha	Silk	31	19.8

1. WRITE QUERY TO RETURN THE EMAIL, FIRST NAME, LAST NAME, & GENRE OF ALL ROCK MUSIC LISTENERS. RETURN YOUR LIST ORDERED ALPHABETICALLY BY EMAIL STARTING WITH A

```
Select Distinct
Customer.Email,
Customer.First_Name,
Customer.Last_Name,
Genre.Name As Genre
From Customer
Join Invoice on Customer.Customer_Id = Invoice.Customer_Id
Join Invoice_Line on Invoice.Invoice_Id = Invoice_Line.Invoice_Id
Join Track on Invoice_Line.Track_Id = Track.Track_Id
Join Genre on Track.Genre_Id = Genre.Genre_Id
Where Genre.Name = 'Rock'
Order By Customer.Email Asc;
```

Email	First_Name	Last_Name	Genre
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	Alexandre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	Bjørn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock

2. LET'S INVITE THE ARTISTS WHO HAVE WRITTEN THE MOST ROCK MUSIC IN OUR DATASET. WRITE A QUERY THAT RETURNS THE ARTIST NAME AND TOTAL TRACK COUNT OF THE TOP 10 ROCK BANDS

```sql
Select
Artist.Name as Artist_Name,
Count(Track.Track_Id) as Total_Rock_Tracks
From Artist
Join Album on Artist.Artist_Id = Album.Artist_Id
Join Track on Album. ALbum_Id = Track.Album_Id
Join Genre on Track.Genre_Id = Genre.Genre_Id
Where Genre.Name = 'Rock'
Group By Artist.Name
Order By Total_Rock_Tracks Desc
Limit 10;
```

Artist_Name	Total_Rock_Tracks
Led Zeppelin	114
U2	112
Deep Purple	92
Iron Maiden	81
Pearl Jam	54

3. RETURN ALL THE TRACK NAMES THAT HAVE A SONG LENGTH LONGER THAN THE AVERAGE SONG LENGTH. RETURN THE NAME AND MILLISECONDS FOR EACH TRACK. ORDER BY THE SONG LENGTH WITH THE LONGEST SONGS LISTED FIRST

```sql
SELECT
    Track.name AS Track_name, milliseconds
FROM
    Track
WHERE
    milliseconds > (SELECT
            AVG(milliseconds)
        FROM
            Track)
ORDER BY milliseconds DESC
```

Track_name	milliseconds
Occupation / Precipice	5286953
Through a Looking Glass	5088838
Greetings from Earth, Pt. 1	2960293
The Man With Nine Lives	2956998
Battlestar Galactica, Pt. 2	2956081
Battlestar Galactica, Pt. 1	2952702

1. FIND HOW MUCH AMOUNT SPENT BY EACH CUSTOMER ON ARTISTS? WRITE A QUERY TO RETURN CUSTOMER NAME, ARTIST NAME AND TOTAL SPENT

```sql
SELECT
    Customer.first_name,
    customer.last_name,
    Artist.Name AS Artist_Name,
    ROUND(SUM(Invoice_Line.Unit_Price * Invoice_Line.Quantity),
            2) AS Total_Spent
FROM
    Customer
        JOIN
    Invoice ON Customer.Customer_Id = Invoice.Customer_Id
        JOIN
    Invoice_Line ON Invoice.Invoice_Id = Invoice_Line.Invoice_Id
        JOIN
    Track ON Invoice_Line.Track_Id = Track.Track_Id
        JOIN
    Album ON Track.Album_Id = Album.Album_Id
        JOIN
    Artist ON Album.Artist_Id = Artist.Artist_Id
GROUP BY Customer.first_name , Customer.last_name , Artist_Name
ORDER BY Total_Spent DESC;
```

first_name	last_name	Artist_Name	Total_Spent
Hugh	O'Reilly	Queen	27.72
Wyatt	Girard	Frank Sinatra	23.76
Helena	Holý	Red Hot Chili Peppers	19.8
Robert	Brown	Creedence Clearwater Revival	19.8
François	Tremblay	The Who	19.8
František	Wichterlová	Kiss	19.8
Aaron	Mitchell	James Brown	19.8
Niklas	Schröder	Queen	18.81
Heather	Leacock	House Of Pain	18.81

2. WE WANT TO FIND OUT THE MOST POPULAR MUSIC GENRE FOR EACH COUNTRY. WE DETERMINE THE MOST POPULAR GENRE AS THE GENRE WITH THE HIGHEST AMOUNT OF PURCHASES. WRITE A QUERY THAT RETURNS EACH COUNTRY ALONG WITH THE TOP GENRE. FOR COUNTRIES WHERE THE MAXIMUM NUMBER OF PURCHASES IS SHARED RETURN ALL GENRES

```sql
With Genre_Purchase_Count As (
    Select
        Customer.Country,
        Genre.Name As Genre,
        Count(*) As Total_Purchases
    From Customer
    Join Invoice On Customer.Customer_Id = Invoice.Customer_Id
    Join Invoice_Line On Invoice.Invoice_Id = Invoice_Line.Invoice_Id
    Join Track On Invoice_Line.Track_Id = Track.Track_Id
    Join Genre On Track.Genre_Id = Genre.Genre_Id
    Group By Customer.Country, Genre.Name
),
Max_Purchases_Per_Country As (
    Select
        Country,
        Max(Total_Purchases) As Max_Purchases
    From Genre_Purchase_Count
    Group By Country
)
```

```sql
    Select
        gpc.country,
        gpc.genre,
        gpc.total_purchases
    From Genre_Purchase_Count gpc
    Join Max_Purchases_Per_Country mpc
      On gpc.country = mpc.country And gpc.total_purchases = mpc.max_purchases
    Order By gpc.country Asc, gpc.genre Asc;
```

country	genre	total_purchases
Argentina	Alternative & Punk	17
Australia	Rock	34
Austria	Rock	40
Belgium	Rock	26
Brazil	Rock	205
Canada	Rock	333
Chile	Rock	61

3. WRITE A QUERY THAT DETERMINES THE CUSTOMER THAT HAS SPENT THE MOST ON MUSIC FOR EACH COUNTRY. WRITE A QUERY THAT RETURNS THE COUNTRY ALONG WITH THE TOP CUSTOMER AND HOW MUCH THEY SPENT. FOR COUNTRIES WHERE THE TOP AMOUNT SPENT IS SHARED, PROVIDE ALL CUSTOMERS WHO SPENT THIS AMOUNT

```sql
With Customer_Total_Spent As (
    Select
        Customer.Country,
        Customer.Customer_Id,
        Customer.First_name, Customer.last_name,
        Round(Sum(Invoice.Total), 2) As Total_Spent
    From Customer
    Join Invoice On Customer.Customer_Id = Invoice.Customer_Id
    Group By Customer.Country, Customer.Customer_Id, Customer.first_name, Customer.last_name
),
Max_Spent_Per_Country As (
Select
        Country,
        Max(Total_Spent) As Max_Spent
    From Customer_Total_Spent
    Group By Country
)
  Select
      cts.country,
      cts.first_name, cts.last_name,
      cts.total_spent
  From Customer_Total_Spent cts
  Join Max_Spent_Per_Country mspc
    On cts.country = mspc.country And cts.total_spent = mspc.max_spent
  Order By cts.country Asc,  cts.first_name, cts.last_name Asc;
```

country	first_name	last_name	total_spent
Argentina	Diego	Gutiérrez	39.6
Australia	Mark	Taylor	81.18
Austria	Astrid	Gruber	69.3
Belgium	Daan	Peeters	60.39
Brazil	Luís	Gonçalves	108.9
Canada	François	Tremblay	99.99
Chile	Luis	Rojas	97.02
Czech Republic	František	Wichterlová	144.54
Denmark	Kara	Nielsen	37.62